# Navigating AI in evidence synthesis

## AI tools across the review workflow

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Map and presentation

lukebax.github.io/ai-review-tools/

# Stages of a systematic review

1 Protocol

2 Search

3 Screening for relevance

4 Quality assessment

5 Data extraction

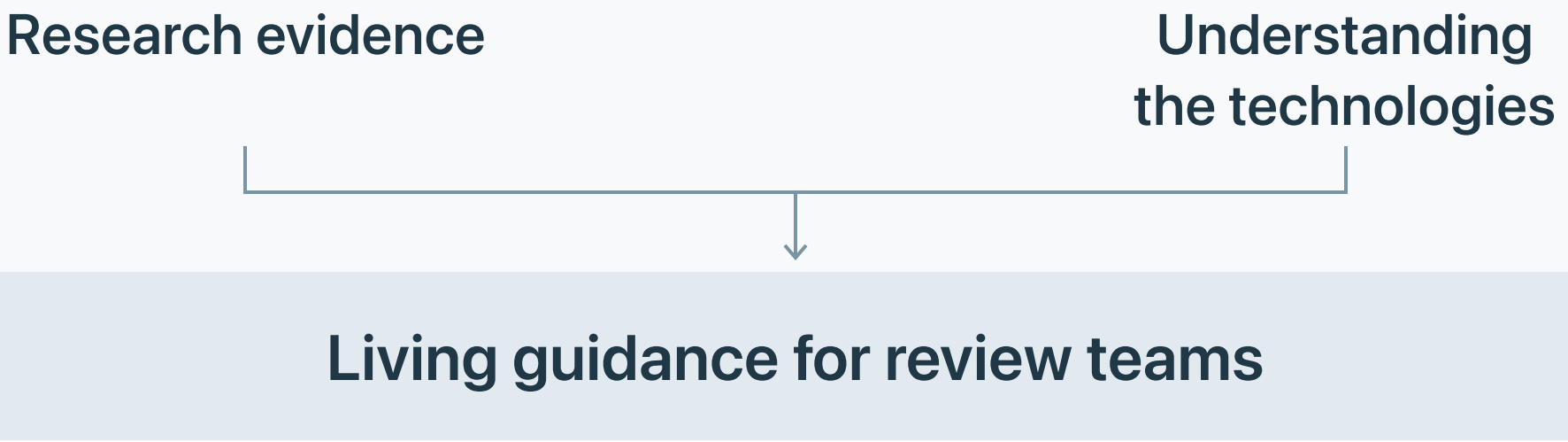
6 Data synthesis

7 Certainty of evidence

8 Manuscript

# RAISE provides guidance for using AI in reviews

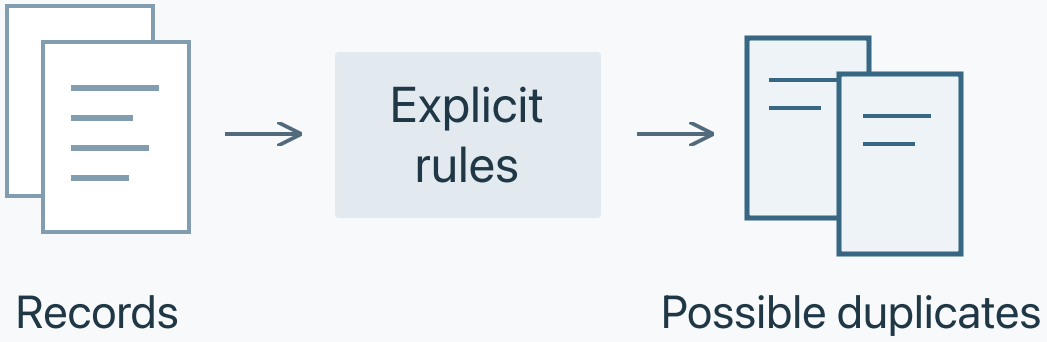
Responsible use of AI in evidence synthesis




Developed by researchers working in evidence synthesis

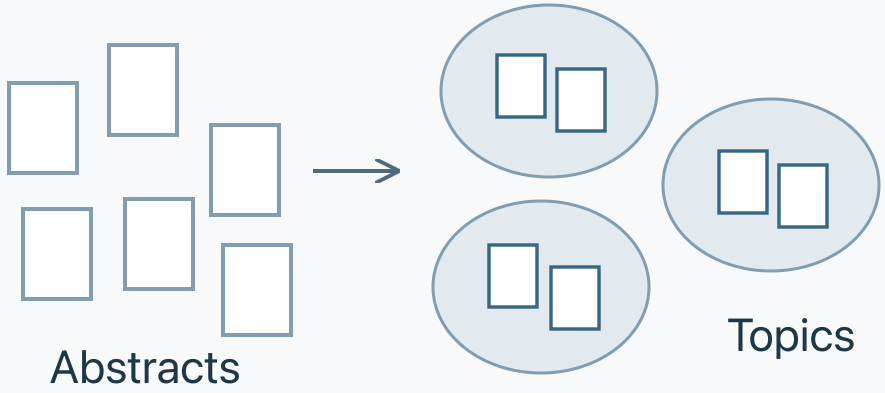
# Explicit rules and patterns in unlabelled data

## Rule-based algorithms




People define the rules used to compare records.

## Unsupervised machine learning

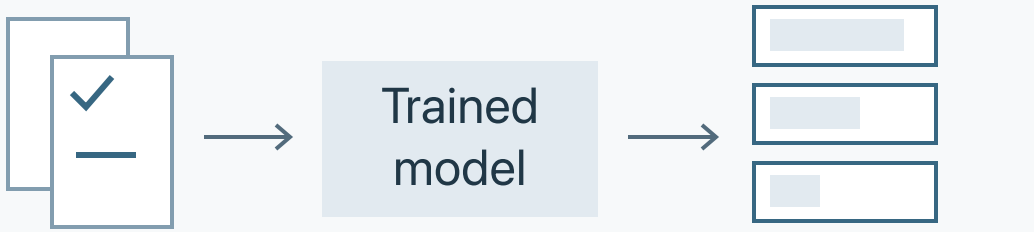



Patterns emerge without supplied labels.

Results depend on the records and the method.

# Learning from labelled examples and generating language

## Supervised machine learning

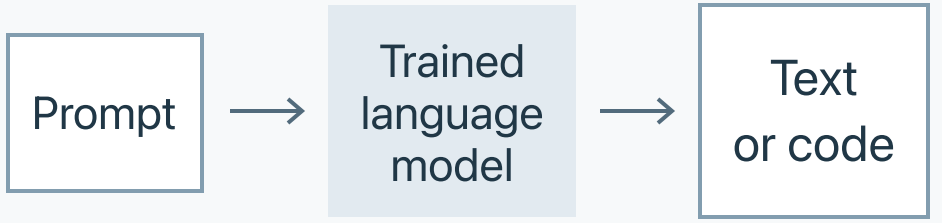



Labels

Relevance ranking

Already trained or trained during the review

## Generative LLMs



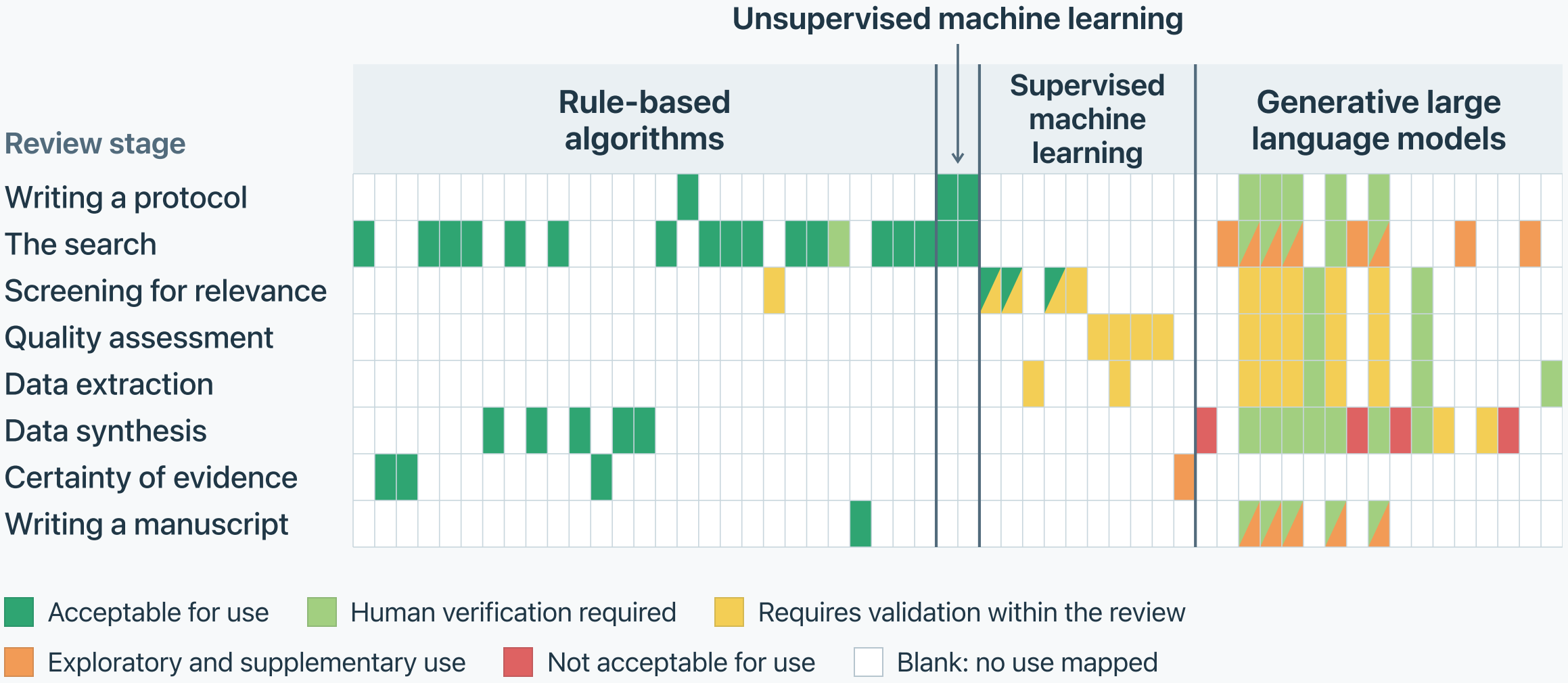

Outputs can vary.
Fluent output can be wrong.

# Recommendations apply to a particular use

| | |
|---|---|
| **Acceptable for use** | Use outputs, accounting for limitations and bias |
| **Human verification required** | Check outputs and correct errors |
| **Requires validation within the review** | Test performance against a prespecified threshold and suitable benchmark |
| **Exploratory and supplementary use** | Explore or supplement established methods |
| **Not acceptable for use** | Do not rely on outputs for this task |

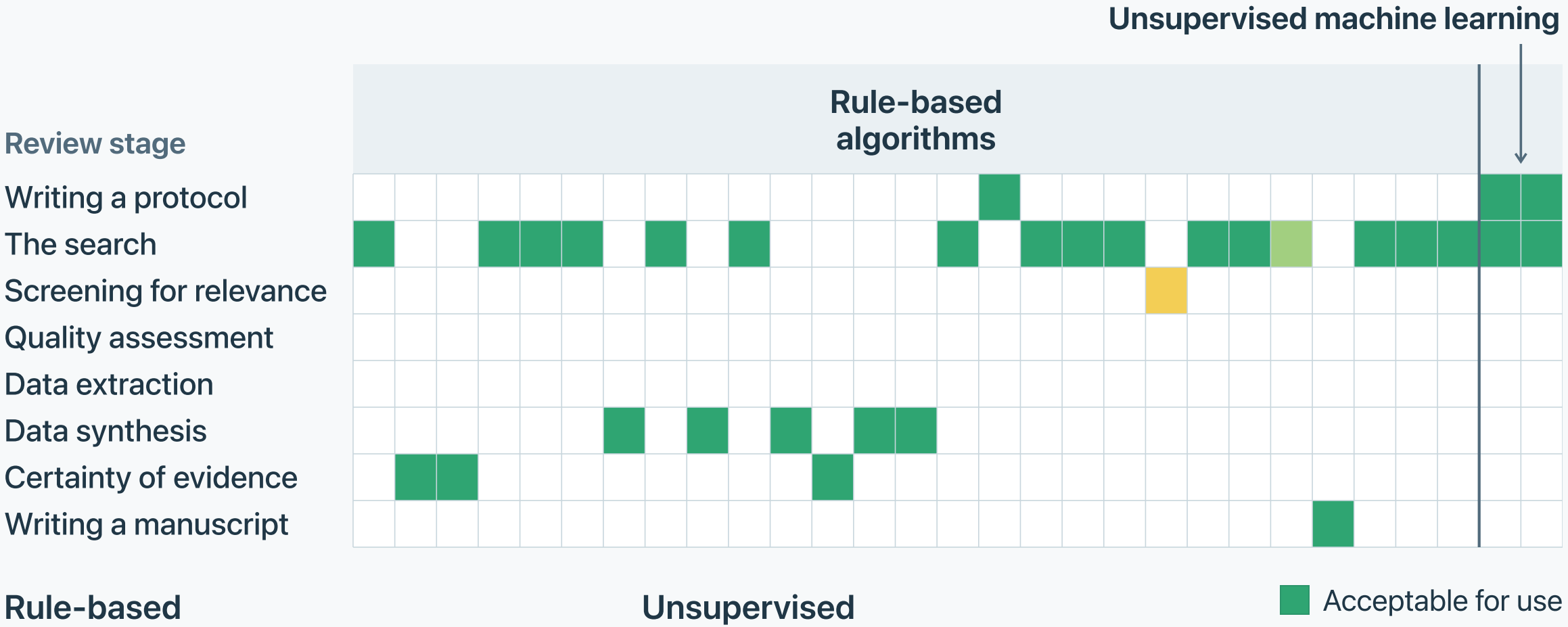
# AI tools across the review stages

Columns are tools. Rows are review stages. Colours show recommendations.



# Most of these mapped uses are acceptable

Enlarged rule-based and unsupervised portions of the map




**Rule-based**

Well-defined, established operations

**Unsupervised**

Exploring the literature

# Stopping early changes reliance on the ranking

**Abstrackr**



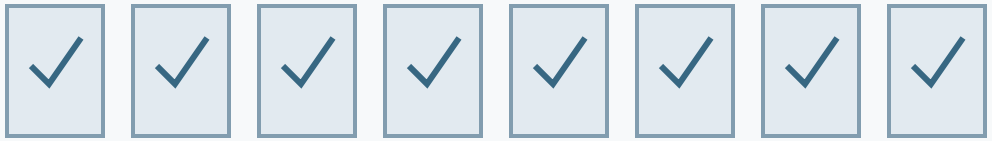
**Covidence**



**EPPI-Reviewer Prioritised Screening**

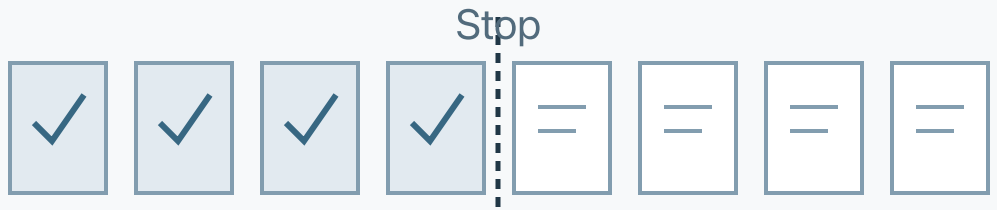


## Screen every record



Every record is screened

## Stop early




Screened

Unscreened

**Acceptable for use**

A poor ranking does not leave records unseen.

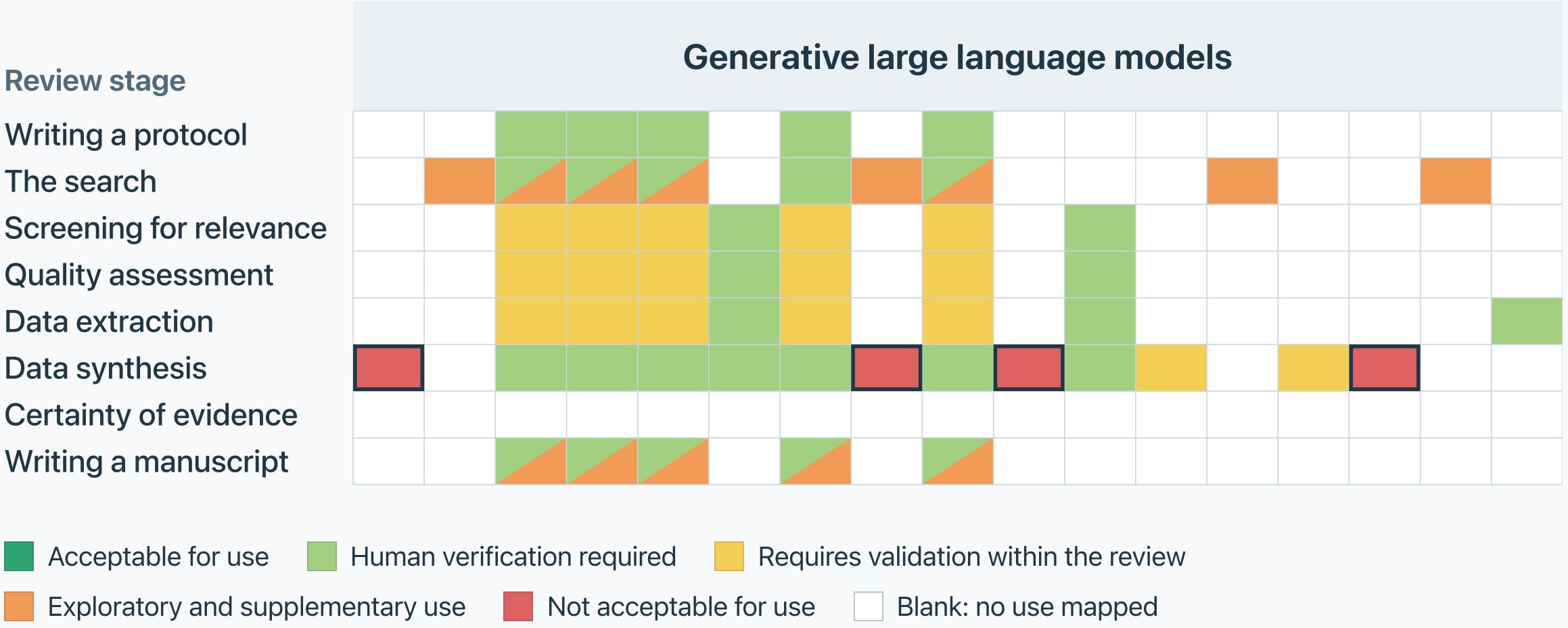
**Requires validation within the review**

Assess the screening and stopping procedure.

The ranking can be identical. What changes is how much we rely on it.

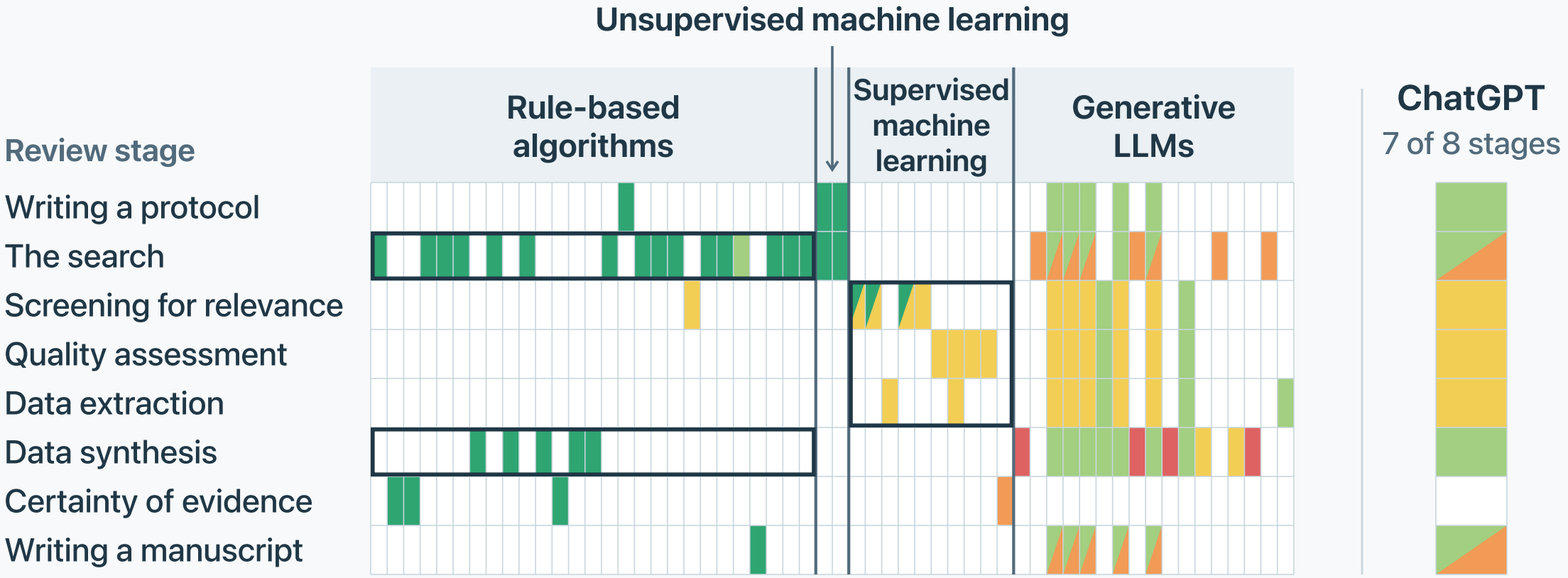
# The mapped LLM uses carry conditions or restrictions

No mapped use is in “Acceptable for use”.



# Tools differ in where and how broadly they are used

Stage clusters in the overview, with ChatGPT enlarged alongside




Rule-based
Search and synthesis

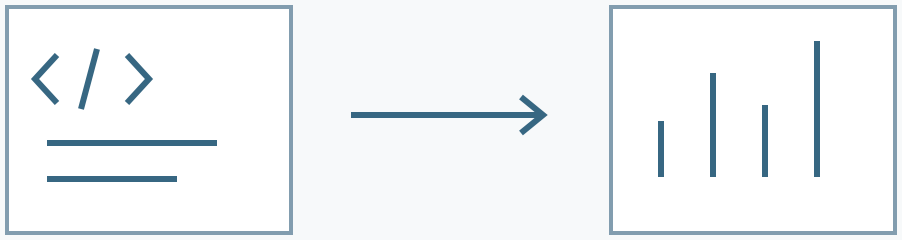
Supervised
Screening, quality assessment, extraction

ChatGPT
Mapped in 7 of 8 stages

# Verification and validation ask different questions

## Verification

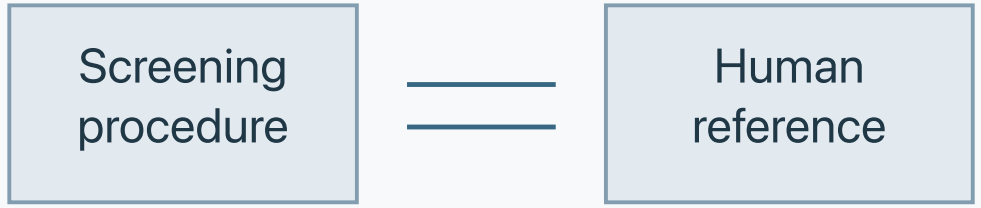
Check the output




Does it carry out the intended analysis?

## Validation

Evaluate performance




Recall: proportion of eligible records found

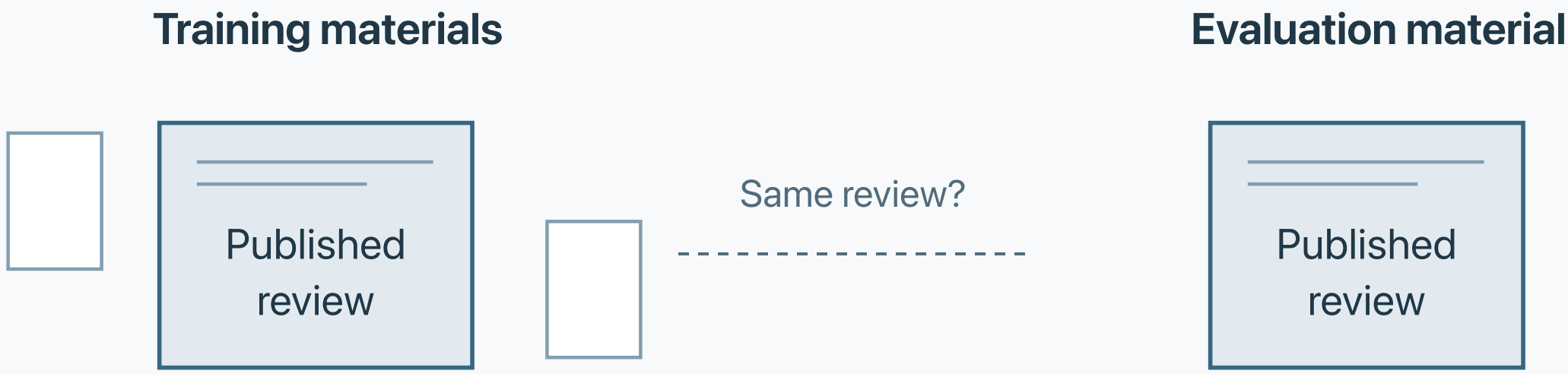
Applicable prior evidence may suffice. Follow any within-review requirement.

Suitable sampling or stopping methods can help, but assessment has a workload cost.

A workflow may need both.

# Evaluation needs independence from development

Possible training–evaluation overlap: a hypothetical example



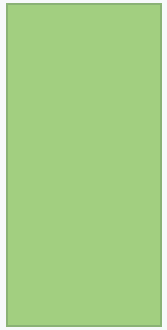

Apparent performance may overstate performance on new material.

Incomplete disclosure makes overlap harder to establish.

# Two different uses during screening

Screening for relevance

## Google Translate

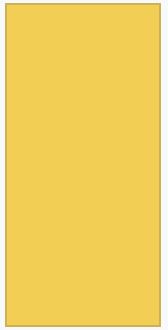


Translation support

**Human verification required**

Check the translation against the original.

## Google’s Gemini



Relevance judgements

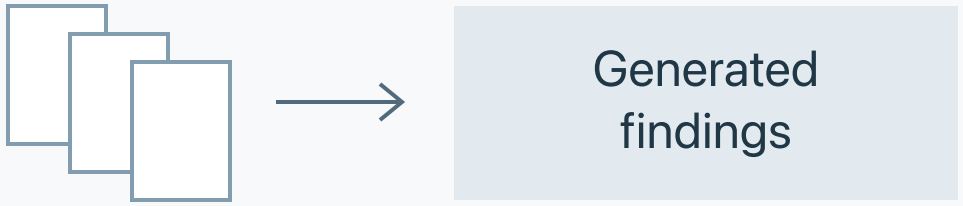
**Requires validation within the review**

Evaluate the screening procedure’s performance.

Selected screening cells from the map.

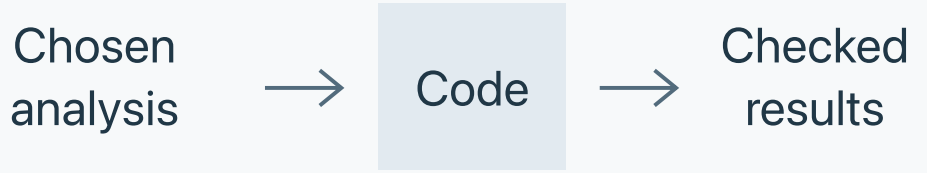
# Synthesising findings differs from drafting analysis code

## LLM-derived synthesis across studies




**Not acceptable for use in a formal review**

## LLM-drafted code for a chosen analysis

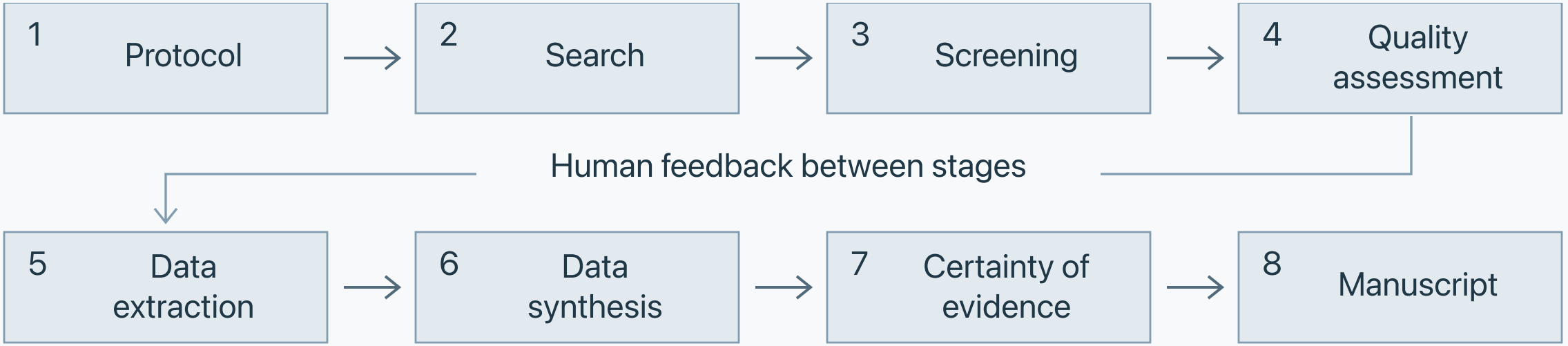



**Human verification required**

Qualitative coding or themes: validation and verification.
The boundary with delegated synthesis is not fully defined.

# RAISE also restricts complete agentic reviews

**Complete agentic review: not acceptable for use**




Human feedback does not remove this restriction.

Individual tasks need separate assessment.

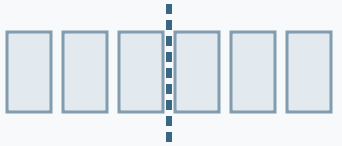
# Is this tool suitable for my review?

Prioritised screening with early stopping

1

## What is the purpose of the tool?

Assess ranking plus the stopping procedure.



2

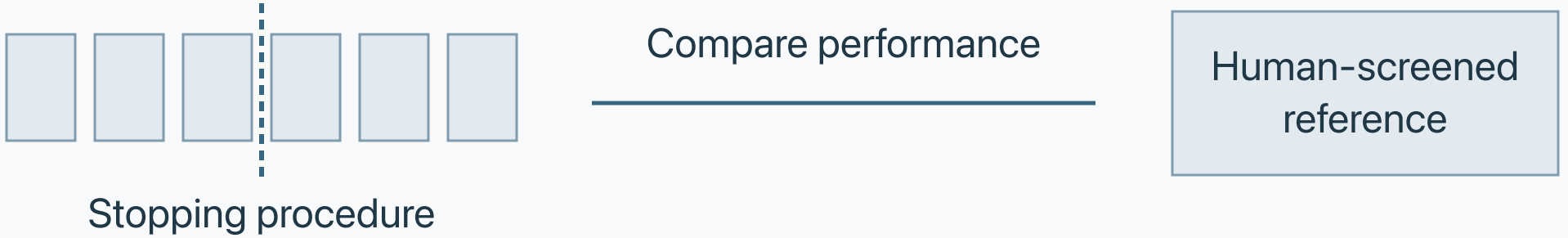
## Where have the training, testing and validation data come from?

Review-team labels: quality and selection.
Method evaluations: relevance to this review.



# 3 Is the tool validated and performing sufficiently well for use?

Prioritised screening with early stopping




Evidence for the stopping procedure

Set required performance in advance.
Establish whether it is met within this review.

An appropriate human-screened reference

# Team, resources and documentation

Prioritised screening with early stopping

4 **Does the team have the expertise and resource to use the AI tool effectively, and is the usability acceptable?**

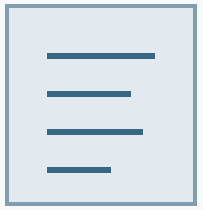
Skills, assessment time and a workable process

5 **Is there sufficient transparency around licenses and other available documentation?**

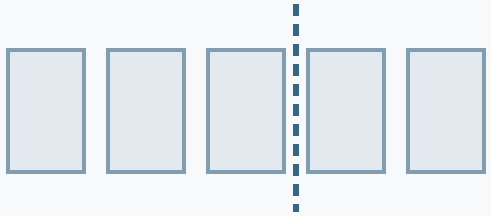
Instructions, access, data handling, version and settings

Missing stopping evidence must be resolved before reliance.

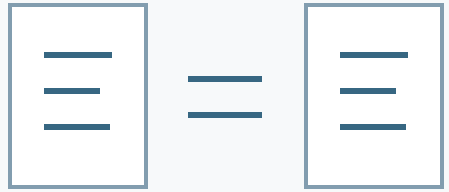
# Choosing a suitable use of AI



Tool and task



Reliance on outputs



Evidence and checks

Map and presentation

**lukebax.github.io/ai-review-tools/**